MATH 3312, SPRING 2026: MIDTERM 1

This exam has 6 (six) printed pages. Please show all your work. Outside materials are NOT permitted for use on this test. Do not spend more than 10 minutes on any one part of a problem. If you get stuck for more than a few minutes, just move on to the next part. *You should feel free to use the results of earlier problems in your solutions to later ones, regardless of whether you have finished them or not.* You have 50 minutes to work on the test. Write your answers in the space provided.

Name ____________________________________

	Score
Problem 1	/10
Problem 2	/10
Problem 3	/10
Problem 4	/10
Problem 5	/10
Total	/50

Name ________________________

(1) (10 points) **True or false. Write one or two lines justifying your answer**:

(a) If R is a finite integral domain, then every ideal in $R[x]$ is principal.

(b) A field of order 5^8 can contain a field of order 5^6.

(c) If $\alpha \in \mathbb{F}_9$ is a zero for $x^2 - x - 1$, then so is α^3.

(d) There exists a finite field k with $k^\times \simeq \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$.

(e) Every ring homomorphism $k \to R$ from a field k to a non-zero ring R is injective.

Name ____________________

(2) (10 points) Give examples of the following or explain why an example *cannot* exist:

(a) A non-zero ideal $J \leq \mathbb{Z}[x]$ such that $\mathbb{Z}[x]/J$ is an integral domain but not a field.

(b) An element of $(\mathbb{F}_3[x]/(x^2+1))^\times$ of order 8.

Name ____________________________________

(3) (10 points)

(a) Compute the number of irreducible polynomials of degree 2 in $\mathbb{F}_5[x]$.

(b) (*Note the change from* 5 *to* 3*!*) Pick two *distinct* irreducible polynomials $f_1(x), f_2(x) \in \mathbb{F}_3[x]$ of degree 2 with leading coefficient 1, and construct an explicit isomorphism of fields $\mathbb{F}_3[x]/(f_1(x)) \xrightarrow{\simeq} \mathbb{F}_3[x]/(f_2(x))$.

Name ________________________________

(4) (10 points)

(a) Show that the following are equivalent for a polynomial $f(x) \in \mathbb{F}_p[x]$ of degree r:

(i) $f(x)$ is irreducible.

(ii) The smallest field in which $f(x)$ admits a zero is $\mathbb{F}_{p^r}$.

(b) Find a prime p such that $x^4 - x^3 + x^2 + x - 1 \in \mathbb{F}_p[x]$ is irreducible.
Hint: $x^5 + 1 = (x+1)(x^4 - x^3 + x^2 - x + 1)$

Name ____________________

(5) (10 points) Construct a commutative ring R with the following property: Giving a ring homomorphism $R \to R'$ to another commutative ring R' is equivalent to giving a pair of elements $(r_1, r_2) \in R' \times R'$ satisfying the equation $r_1^3 - 2r_2 + 1 = 0$.